

Price Discrimination and Monopoly: Nonlinear Pricing

Introduction

- **Annual subscriptions generally cost less in total than one-off purchases**
- **Buying in bulk usually offers a price discount**
 - these are price discrimination reflecting quantity discounts
 - prices are *nonlinear*, with the unit price dependent upon the quantity bought
 - allows pricing nearer to willingness to pay
 - so should be more profitable than third-degree price discrimination
- **How to design such pricing schemes?**
 - depends upon the information available to the seller about buyers
 - distinguish *first-degree* (personalized) and *second-degree* (menu) pricing

First-degree price discrimination (cont'd)

- **Monopolist can charge maximum price that each consumer is willing to pay**
- **Extracts *all* consumer surplus**
- **Since profit is now total surplus, find that first-degree price discrimination is *efficient***

First-degree price discrimination (cont'd)

- **Suppose that you own five antique cars**
- **Market research shows there are collectors of different types**
 - **keenest is willing to pay \$10,000 for a car, second keenest \$8,000, third keenest \$6,000, fourth keenest \$4,000, fifth keenest \$2,000**
 - **sell the first car at \$10,000**
 - **sell the second car at \$8,000**
 - **sell the third car to at \$6,000 and so on**
 - **total revenue \$30,000**
- **Contrast with linear pricing: all cars sold at the same price**
 - **set a price of \$6,000**
 - **sell three cars**
 - **total revenue \$18,000**

First-degree price discrimination (cont'd)

- **First-degree price discrimination is highly profitable but requires**
 - detailed information
 - ability to avoid arbitrage
- **Leads to the efficient choice of output: since price equals marginal revenue and $MR = MC$**
 - no value-creating exchanges are missed

First-degree price discrimination (cont'd)

- **The information requirements appear to be insurmountable**
 - but not in particular cases
 - tax accountants, doctors, students applying to private universities
- **No arbitrage is less restrictive but potentially a problem**
- **In the antique cars example, each buyer purchased one car; let's now consider a case where buyers buy multiple products.**
- **We consider two pricing schemes that will achieve the same outcome as in the antique cars example**
 - non-linear prices
 - *two-part pricing* as a particular example of non-linear prices
 - charge a quantity-independent fee (e.g. membership) plus a per unit usage charge
 - *block pricing* is another
 - bundle total charge and quantity in a package

Two-part pricing

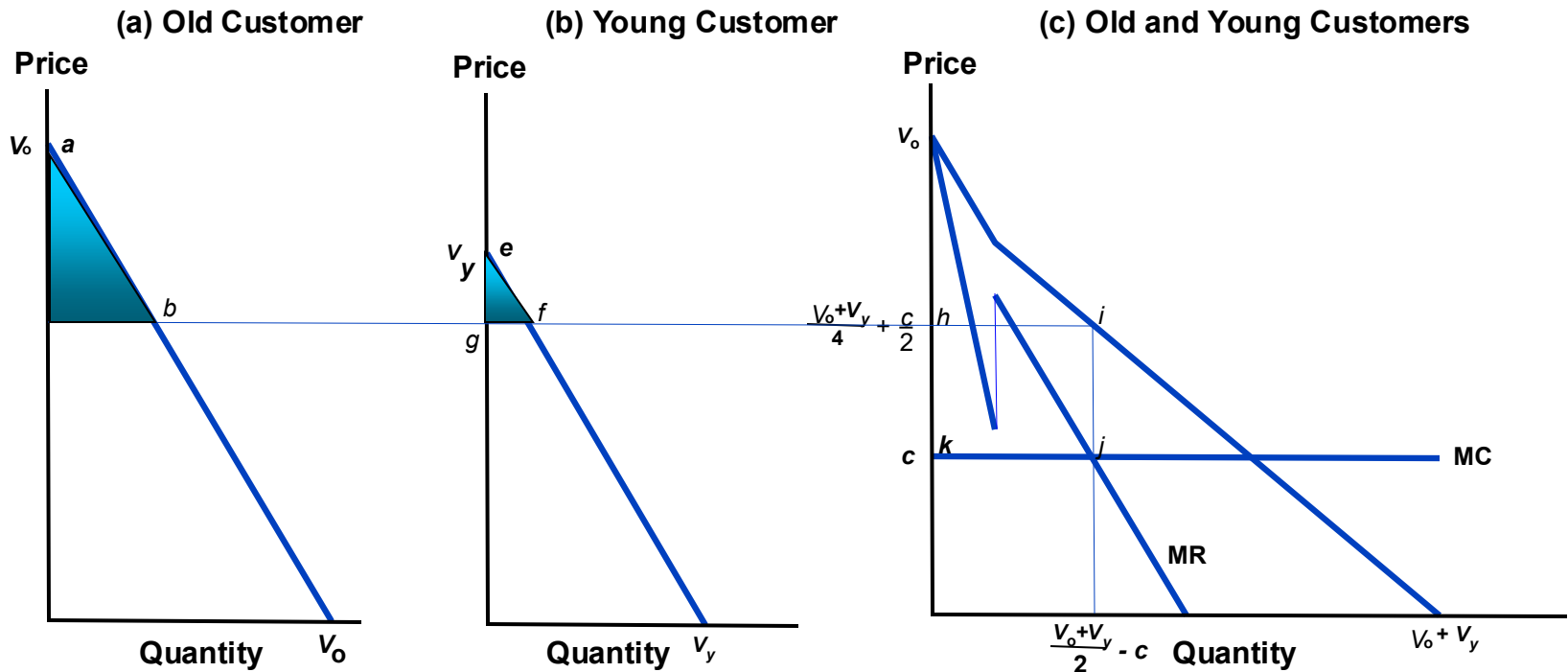
- **A simple example: only two customers (old and young)**
- **Amusement park sells tickets to the old and young customers**
 - **Old: demand for entry plus Q_o rides is $P = V_o - Q_o$**
 - **Young: demand for entry plus Q_y rides is $P = V_y - Q_y$**
 - **Again: one for each type**
 - **Assume that $V_o > V_y$: Old is willing to pay more than Young**
 - **Cost of operating the amusement park:
 $C(Q) = cQ$, where $Q = Q_o + Q_y$**
- **Demand and costs are all in daily units**
- **What are the model risks here?**

Two-part pricing (cont'd)

- **Suppose that the park operator applies “traditional” linear pricing: free entry and a set price for ride tickets**
 - aggregate demand is $Q = Q_o + Q_y = (V_o + V_y) - 2P$
 - invert to give: $P = (V_o + V_y)/2 - Q/2$
 - MR is then $MR = (V_o + V_y)/2 - Q$
 - equate MR and MC, where $MC = c$ and solve for Q to give $Q_U = (V_o + V_y)/2 - c$
 - substitute into aggregate demand to give the equilibrium price $P_U = (V_o + V_y)/4 + c/2$
 - the Old customer buys $Q_o = (3V_o - V_y)/4 - c/2$ ride tickets
 - the Young customer buys $Q_y = (3V_y - V_o)/4 - c/2$ ride tickets
 - profit from the Old and Young is $\pi_U = (V_o + V_y - 2c)^2 / 8$

Two-part pricing (cont'd)

This example can be illustrated as follows:



Linear pricing leaves each type of consumer with consumer surplus

Two-part pricing (cont'd)

- **Park operator can do better than this**
- **Consumer surplus at the uniform linear price is:**
 - Old: $CS_o = (V_o - P_U) \cdot Q_o / 2 = (Q_o)^2 / 2$
 - Young: $CS_y = (V_y - P_U) \cdot Q_y / 2 = (Q_y)^2 / 2$
- **So, charge an entry fee (just less than):**
 - $E_o = CS_o$ to each Old customer and $E_y = CS_y$ to each Young customer
 - *check IDs to implement this policy*
 - each type will still be willing to enter the amusement park and buy the equilibrium number of ride tickets
- **So, this increases profit by E_o for the Old and E_y for the Young customer**

Two-part pricing (cont'd)

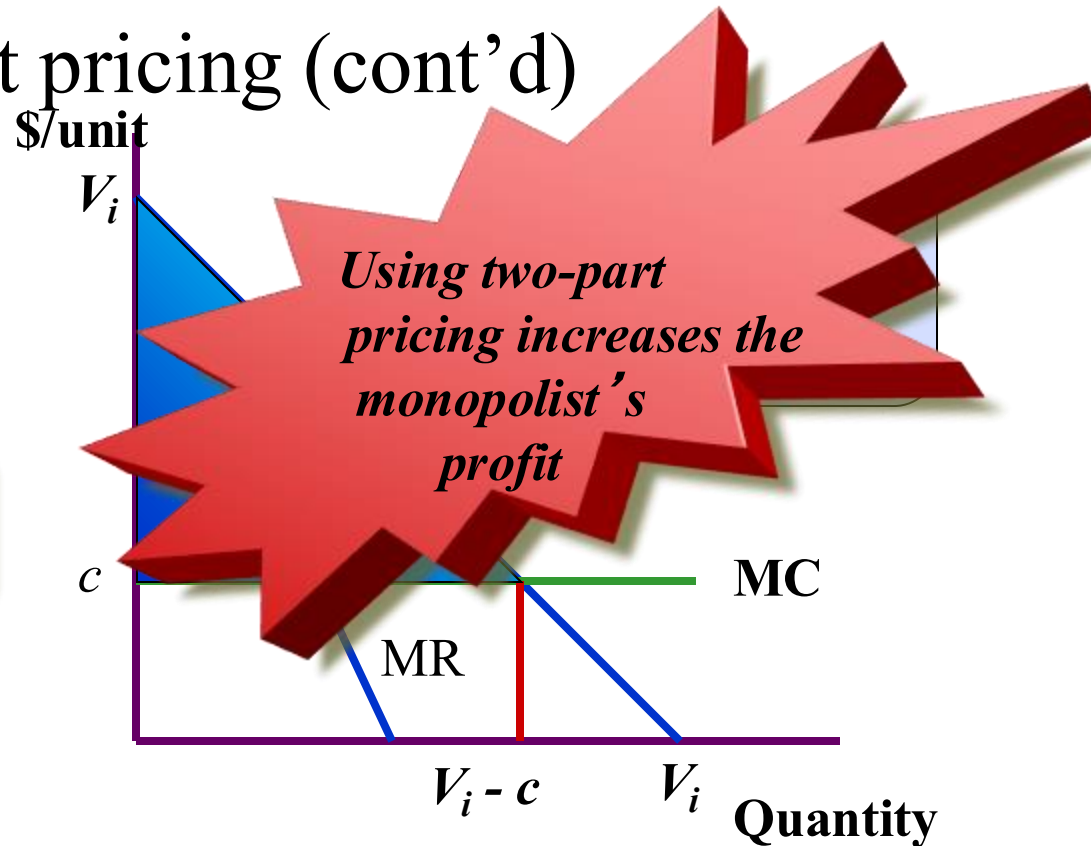
- **The park operator can do even better**
 - reduce the price per ride
 - this increases consumer surplus
 - but the additional consumer surplus can be extracted through a higher entry fee
- **Consider the best that the amusement park operator can do with respect to each type of consumer**

Two-Part pricing (cont'd)

Set the unit price equal to marginal cost

This gives consumer surplus of $(V_i - c)^2/2$

Set the entry charge to $(V_i - c)^2/2$

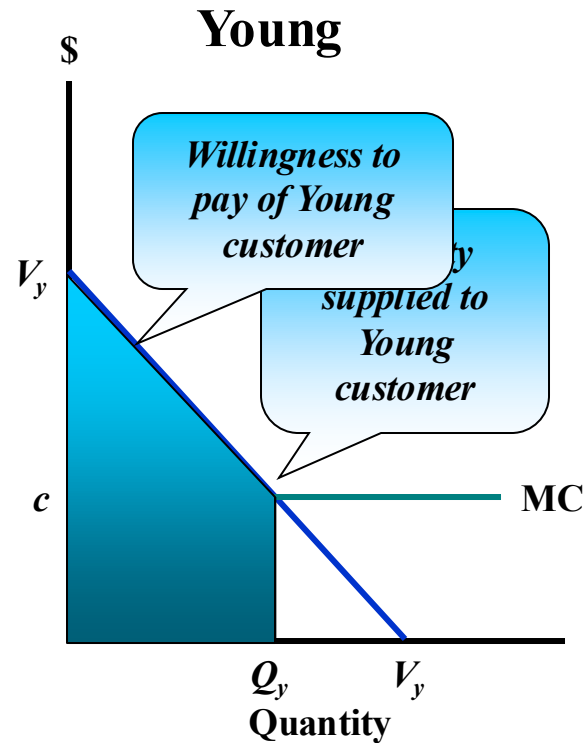
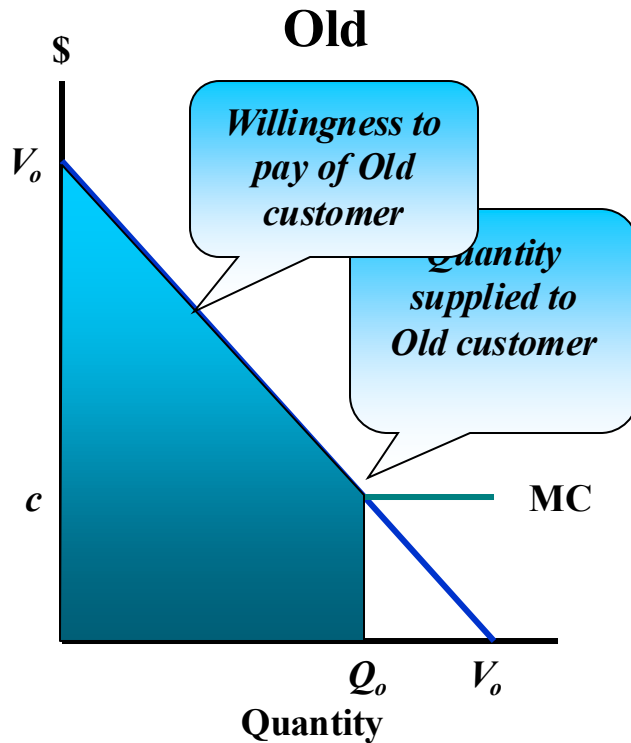


Profit from the Old and Young now $\pi_d = [(V_o - c)^2 + (V_y - c)^2]/2$

Block pricing

- **There is another pricing method that the park operator can apply**
 - offer a *package* of “Entry plus X rides for \$Y”
- **To maximize profit apply two rules**
 - set the quantity offered to each consumer type equal to the amount that type would buy at price equal to marginal cost
 - set the total charge for each consumer type to the total willingness to pay for the relevant quantity
- **Return to the example...**

Block pricing (cont'd)



$$WTP_o = (V_o - c)^2/2 + (V_o - c)c = (V_o^2 - c^2)/2$$

$$WTP_y = (V_y - c)^2/2 + (V_y - c)c = (V_y^2 - c^2)/2$$

Block pricing (cont'd)

- How to implement this policy?
 - Check ID at the door
 - give guests punch-cards based on the number of rides
- What could go wrong?

A final comment

- **One final point**

- **average price that is paid by the Old customer:**
 $(V_o^2 - c^2)/2(V_o - c) = (V_o + c)/2$
- **average price paid by the Young customer:**
 $(V_y^2 - c^2)/2(V_o - c) = (V_y + c)/2$
- **identical to the third-degree price discrimination (linear) prices**
- **but the profit outcome is much better with first-degree price discrimination. Why?**
 - *consumer equates MC of last unit bought with marginal benefit*
 - *with linear pricing MC = AC (= average price)*
 - *with first-degree price discrimination MC of last unit bought is less than AC (= average price) so more is bought*

Second-degree price discrimination

- **What if the seller cannot distinguish between buyers?**
 - perhaps they differ in income (unobservable)
- **Then the type of price discrimination just discussed is impossible**
- **High-income buyer will pretend to be a low-income buyer**
 - to avoid the high entry price
 - to pay the smaller total charge
- **Take a specific example**
 - $P_h = 16 - Q_h$
 - $P_l = 12 - Q_l$
 - $MC = 4$
 - Many customers, but each of them either high type or low type
- **Once again, remember the model risks!**

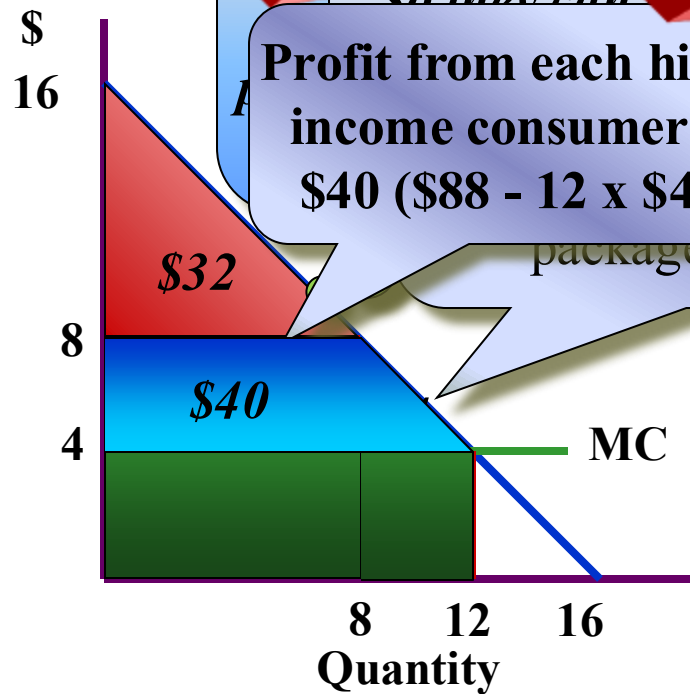
Second-degree price discrimination (cont'd)

- **First-degree price discrimination requires:**
 - **High Income:** entry fee \$72 and \$4 per ride or entry plus 12 rides for a total charge of \$120
 - **Low Income:** entry fee \$32 and \$4 per ride or entry plus 8 rides for total charge of \$64
- **This will not work**
 - high income types get no consumer surplus from the package designed for them but get consumer surplus from the other package
 - So, they will pretend to be low income even if this limits the number of rides they can buy
- **Need to design a “menu” of offerings targeted at the two types**

Second-degree price discrimination (cont'd)

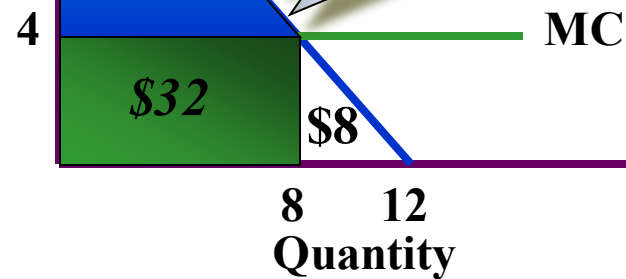
- **The seller has to compromise**
- **Design a pricing scheme that makes buyers**
 - *reveal their true types*
 - *self-select* the quantity/price package designed for them
- **Essence of *second-degree price discrimination***
- **It is “like” first-degree price discrimination**
 - the seller knows that there are buyers of different types
 - *but* the seller is not able to identify the different types
- **A two-part tariff is ineffective**
 - allows deception by buyers
- **Use *quantity discounting***
- **Back to the example...**

Second degree price discrimination (cont'd)



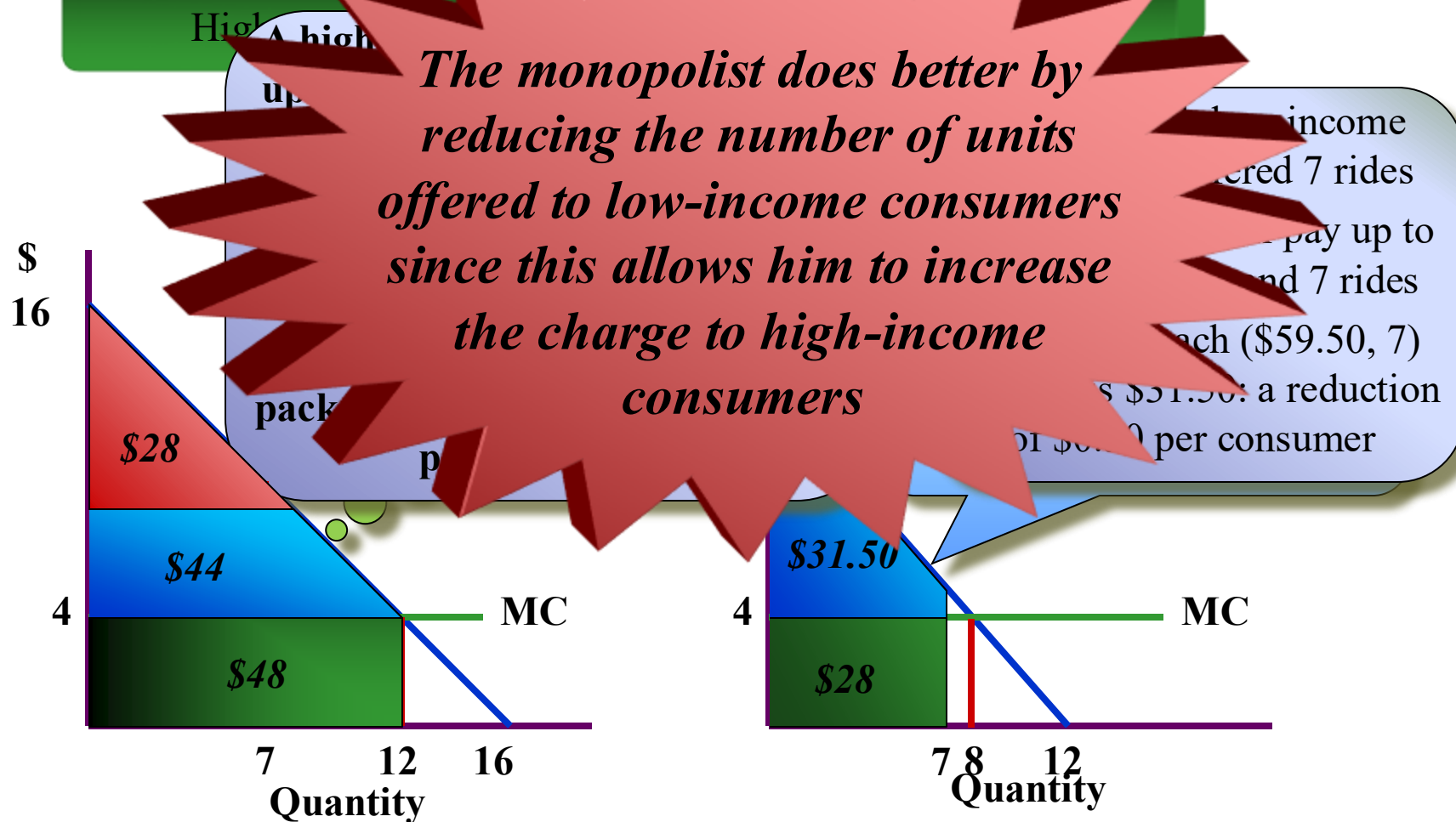
Profit from each high income consumer is \$40 ($\$88 - 12 \times \4)

These packages exhibit quantity discounting: high-income pay \$7.33 per unit and low-income pay \$8



Profit from each low-income consumer is \$32 ($\$64 - 8 \times \4)

Second degree price discrimination (cont'd)



Second-degree price discrimination (cont'd)

- **Will the monopolist always want to supply both types of consumer?**
- **There are cases where it is better to supply only high-demand types**
 - high-class restaurants
 - golf and country clubs
- **Take our example again**
 - suppose that there are N_l low-income consumers
 - and N_h high-income consumers

Second-degree price discrimination (cont'd)

- **Suppose both types of consumer are served**
 - two packages are offered (\$57.50, 7) aimed at low-income and (\$92, 12) aimed at high-income
 - profit is $\$31.50 \times N_l + \$44 \times N_h$
- **Now suppose only high-income consumers are served**
 - then a (\$120, 12) package can be offered
 - profit is $\$72 \times N_h$
- **Is it profitable to serve both types?**
 - Only if $\$31.50 \times N_l + \$44 \times N_h > \$72 \times N_h \Rightarrow 31.50 N_l > 28 N_h$

This requires that $\frac{N_h}{N_l} < \frac{31.50}{28} = 1.125$

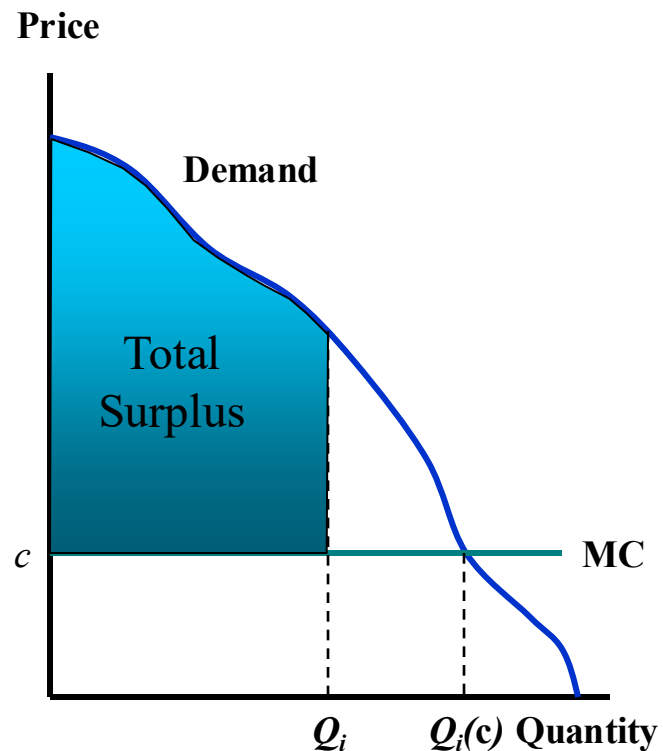
There should not be “too high” a fraction of high-demand consumers

Second-degree price discrimination (cont'd)

- **Characteristics of second-degree price discrimination**
 - extract all consumer surplus from the lowest-demand group
 - leave some consumer surplus for other groups
 - *the incentive compatibility constraint*
 - offer less than the socially efficient quantity to all groups other than the highest-demand group
 - offer quantity-discounting
- **Second-degree price discrimination converts consumer surplus into profit less effectively than first-degree**
- **Some consumer surplus is left “on the table” in order to induce high-demand groups to buy large quantities**

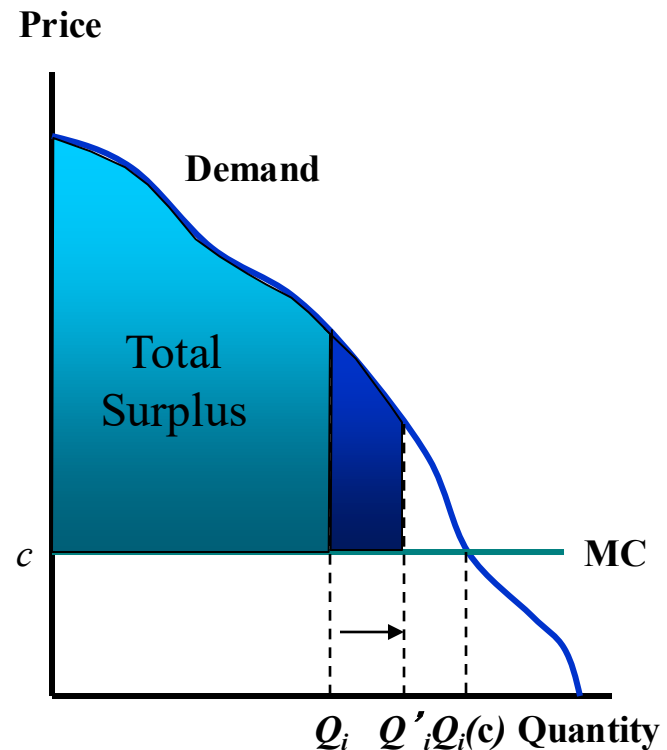
Non-linear pricing and welfare

- **Non-linear price discrimination raises profit**
- **Does it increase social welfare?**
 - *suppose that inverse demand of consumer group i is $P = P_i(Q)$*
 - *marginal cost is constant at $MC = c$*
 - *suppose quantity offered to consumer group i is Q_i*
 - *total surplus – consumer surplus plus profit – is the area between the inverse demand and marginal cost up to quantity Q_i*



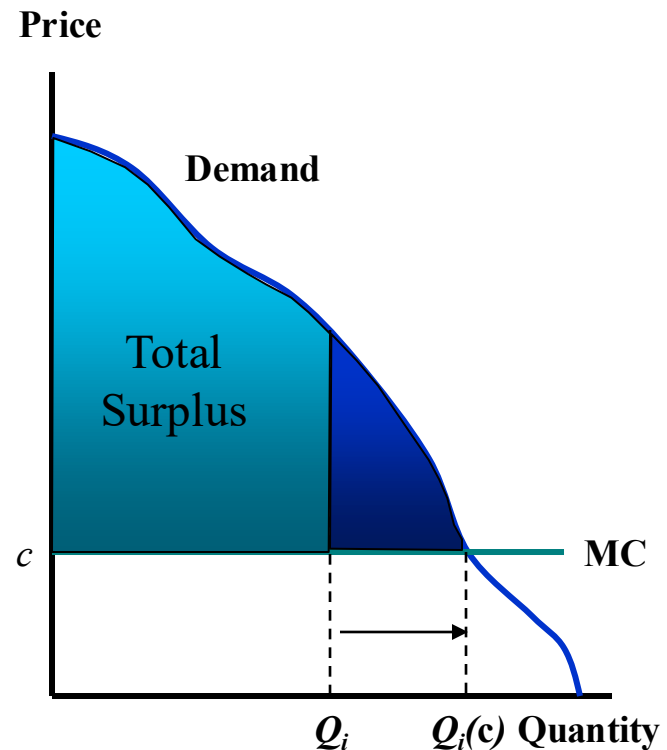
Non-linear pricing and welfare (cont'd)

- Pricing policy affects
 - *distribution of surplus*
 - *output of the firm*
- First is welfare neutral
- Second affects welfare
- Does it increase social welfare?
- Price discrimination increases social welfare of group i if it increases quantity supplied to group i



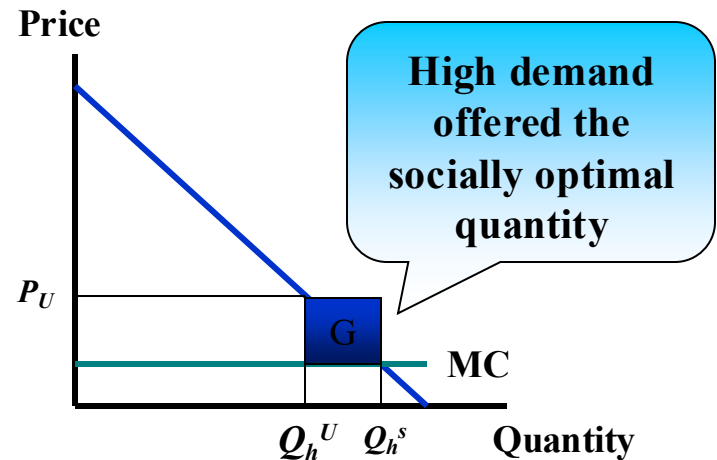
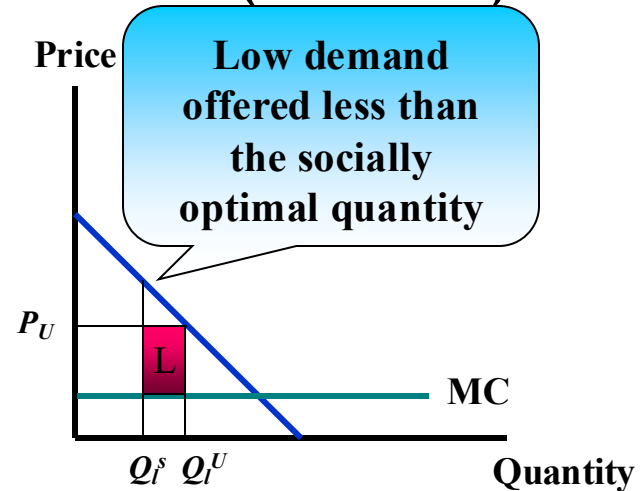
Non-linear pricing and welfare (cont'd)

- **First-degree price discrimination always increases social welfare**
 - extracts all consumer surplus
 - but generates socially optimal output
 - output to group i is $Q_i(c)$
 - this exceeds output with uniform (non-discriminatory) pricing



Non-linear pricing and welfare (cont'd)

- **Menu pricing is less straightforward**
 - suppose that there are two markets
 - low demand
 - high demand
- **Uniform price is P_U**
- **Menu pricing gives quantities Q_l^s, Q_h^s**
- **Welfare loss is greater than L**
- **Welfare gain is less than G**



Non-linear pricing and welfare (cont'd)

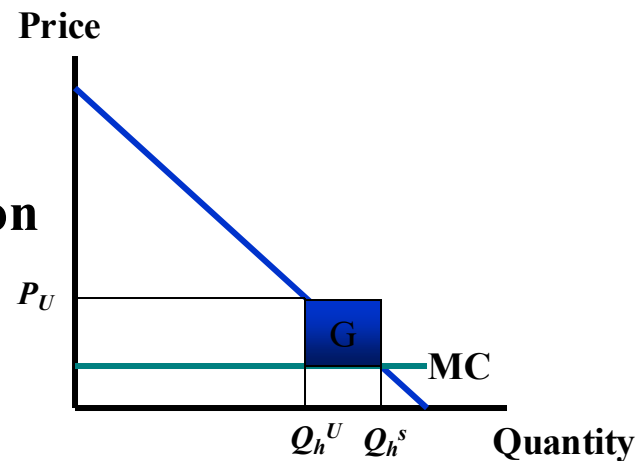
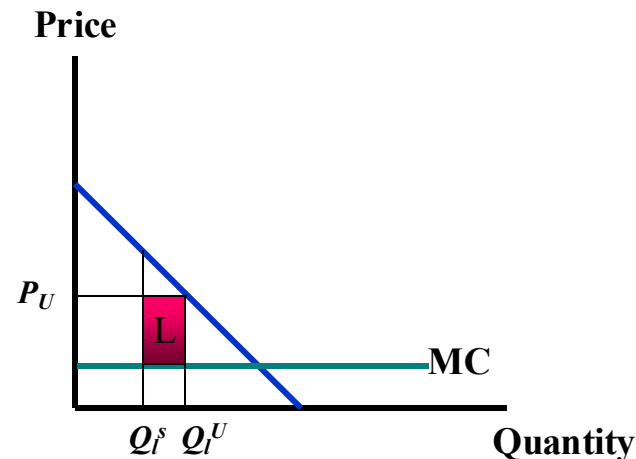
- It follows that

$$\Delta W \leq G - L$$

$$= (P_U - MC)\Delta Q_1 + (P_U - MC)\Delta Q_2$$

$$= (P_U - MC)(\Delta Q_1 + \Delta Q_2)$$

- A necessary condition for second-degree price discrimination to increase social welfare is that it increases total output
- “Like” third-degree price discrimination
- But second-degree price discrimination is more likely to increase output



The incentive compatibility constraint

- *Any offer made to high demand consumers must offer them as much consumer surplus as they would get from an offer designed for low-demand consumers.*
- This is a common phenomenon
 - performance bonuses must encourage effort
 - insurance policies need large deductibles to deter cheating
 - piece rates in factories have to be accompanied by strict quality inspection
 - encouragement to buy in bulk must offer a price discount

